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***h*-index: 5, *i10*-index: 2**

RONTU DAS

C/o Dr. Debashis Kundu
Senior research fellow in Chemical Engineering,
ICT Marathwada Campus, Jalna
Address: 66/111 P. Road Belgachia Howrah.
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Curriculum Vitae

Present Position

Presently working as a Research Associate in Chemical Engineering at the Institute of Chemical Technology Marathwada Campus, Jalna

Career Objectives

To contribute to a cutting-edge computational materials research program as a lead researcher or postdoctoral fellow. Aiming to leverage my Ph.D. research and CSIR-SRF experience in AI-driven modelling and Molecular Dynamics to pioneer new methodologies in material design and analysis.

Ph.D. thesis under the supervision of Dr. Debashis Kundu
Title of the thesis: Artificial Intelligence Assisted Molecular Dynamics Simulation of Carboxymethyl Cellulose–Polyvinyl Alcohol Click Hydrogels for Various Applications
Master of Technology thesis under the supervision of Dr. Debajyoti Goswami
Title of the thesis: Biocatalytic Production of Essential Fatty Acids in Ionic Liquid/Surfactant Mixed System

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1.	**Title**	**NAMoSTe (Novel Atomic & Molecular Structure Explorer)**		
	Application Number	29693/2024-CO/SW	**Certificate No.**	SW-20338/2025
	Date of Filing	23-09-2024	**Date of Granted**	04-04-2025
2.	**Title**	**Swelling analysis for Materials (SaMs)**		
	Application Number	6058/2025-CO/SW	**Certificate No.**	SW-2025021043
	Date of Filing	21-02-2025	**Date of Granted**	23-07-2025
3.	**Title**	**Ionic Conductivity Simulator (IConS)**		
	Application Number	6063/2025-CO/SW	**Certificate No.**	SW-2025021136
	Date of Filing	21-02-2025	**Date of Granted**	04-08-2025

Key Research Areas: Thermodynamics, Molecular Dynamics Simulation, Quantum Chemical Calculation, Data-Driven Neural Network, Physics-Informed Neural Network, Polymeric gel, Material Discovery

Research Publications:

Refereed Journals	
2026	
13	***Rontu Das***, Debashis Kundu. Bridging Atomistic Simulations and Reservoir Computing for Predicting

	Structural and Transport Properties of Thiol–Ene Click-Cross-Linked Carboxymethyl Cellulose Hydrogels. **The Journal of Physical Chemistry B** 130(11) (2026) 3167-3186. DOI: 10.1021/acs.jpcb.5c08098
12.	***Rontu Das***, Debashis Kundu. Understanding the Molecular Level Interaction of Lithium-ions with Cellulose Based Eutectogel: A Molecular Dynamics Study. **Advanced Theory and Simulations** (2026). DOI: 10.1002/adts.202501934
11.	***Rontu Das***, Debashis Kundu. Molecular dynamics simulation of thiol-ene 'click' copolymerized and blended carboxymethyl cellulose and 4-arm polyethylene glycol hydrogel used as ciprofloxacin drug carrier. **Chemical Engineering Science** 324 (2026) 123356. DOI: 10.1016/j.ces.2026.123356
	2025
10.	Ayush S. Gulhane, ***Rontu Das***, Navnath T. Hatvate and Debashis Kundu. Optimizing Solvent Systems for Effective Lutein Solvation: A Molecular Dynamics Approach Using 2-Methyl Tetrahydrofuran, Deep Eutectic Solvent, Microemulsion, and Hexane. **Macromolecular Theory and Simulations (2025).** DOI:10.1002/mats.202500016
9.	***Rontu Das*** and Debashis Kundu. Adsorption of Methylene Blue Dye in the Thiol-Norbornene Crosslinked Polyvinyl Alcohol Hydrogel Surface: A Molecular Dynamics Study. **Environmental Science and Pollution Research** (2025). DOI: 10.1007/s11356-025-36309-3
8.	Ritesh K. Patil, ***Rontu Das***, Debashis Kundu. Structure and Dynamics of Anacardic Acid in Hexane, Ethanol and Carbon Tetrachloride: A Molecular Dynamics Investigation. **Journal of Molecular Modelling** (2025) 31, 107. DOI: 10.1007/s00894-025-06320-z
7.	Abhinav Tidke, ***Rontu Das***, Debashis Kundu. Unveiling Molecular Interactions in Chocolate by Molecular Dynamics: A Study on Caffeine, Fatty Acids and Sugars. **Journal of Chemical & Engineering Data** (2025). DOI: 10.1021/acs.jced.4c00665
6.	***Rontu Das***, Debashis Kundu. Enlightenment the Dynamic Behavior of Norbornene–Modified 'Click' 4–arm Polyethylene Glycol Hydrogel: Delving into Framework Properties and Transport Properties through Molecular Dynamics Simulations. **Computational Materials Science** (2025) 247, 113516. DOI: 10.1016/j.commatsci.2024.113516
	2024
5.	***Rontu Das*** and Debashis Kundu. Understanding the Supramolecular Chemistry of Lutein with Gelatin, Octenyl Succinic Anhydride-Modified Starch Polymers in Simulated Gastro-Intestinal Buffers: A Molecular Dynamics Study. **Macromolecular Chemistry and Physics** (2024) 2400295. DOI: 10.1002/macp.202400295
4.	Abhinav Tidke, ***Rontu Das***, Uday S. Annapure, Debashis Kundu. Solvent Optimization for Caffeine and Tannic Acid Extraction from Guarana Seed: Insights from Molecular Dynamics Simulations. **Journal of Molecular Liquids** (2024) 416, 126462. DOI: 10.1016/j.molliq.2024.126462
3.	Aradhana V. Thombre, ***Rontu Das***, Debashis Kundu. Lithium Ion Transportation in Lignin Infused

	Polyvinyl Alcohol Based Eutectogel: A Molecular Dynamics Framework. **Journal of Energy Storage** (2024) 102, 114060. DOI: 10.1016/j.est.2024.114060
2.	Shreyash P. Deshmukh, ***Rontu Das***, Debashis Kundu. Unraveling the underlying structural & transport mechanism of lithium-ion within Lithium bis(trifluoromethanesulfonyl)imide subjected to organic & inorganic matrix based Eutectogel. **Journal of Power Sources** (2024) 600, 234270. DOI: 10.1016/j.jpowsour.2024.234270
	2023
1.	***Rontu Das***, Debashis Kundu. Structural and Transport Properties of Norbornene-Functionalized Poly(vinyl alcohol) "Click" Hydrogel: A Molecular Dynamics Study, **ACS Sustainable Chemistry & Engineering** (2023) 11(29) 10812–10824. DOI: 10.1021/acssuschemeng.3c01948

Academic Records

Ph.D. in Chemical Engineering	Institute of Chemical Technology	Awarded	2021 – 2026
Master of Technology in Chemical Engineering (Specialization: Petroleum Refinery Engineering)	University of Calcutta	8.07 (CGPA)	2019 – 21
Bachelor of Chemical Engineering	Maulana Abul Kalam Azad University of Technology, West Bengal	8.74 (DGPA)	2015 – 19
12th Std.	Bantra Madhusudan Palchowdhury High School, West Bengal	78.00 %	2013 – 15
10th Std.	Bantra Madhusudan Palchowdhury High School, West Bengal	76.00 %	2013

Achievements

- Awarded the highly competitive **CSIR-Senior Research Fellowship (Direct)** by the Council of Scientific & Industrial Research (May 2024).
- **Demonstrated** strong technical aptitude by successfully qualifying the Graduate Aptitude Test in Engineering (GATE) in Chemical Engineering (2020).

Research Experiences

- **CSIR-Senior Research Fellowship (Direct) | May 2024 – Apr 2026**
    - **Project:** Amalgamation of Molecular Dynamics and Machine Learning Strategies for the Creation of Supersorbent Thiol-ene 'Click' Hydrogel
    - **Research Focus:** AI Assisted Molecular Dynamics Simulation for thiol-ene 'click hydrogel for predicting long term trajectory
- **Junior Research Fellow | SERB-Sponsored Project | Dec 2023 – Apr 2024**

- **Project:** Artificial Intelligence and Molecular Dynamics Driven Modeling Platform for 'Click' Hydrogel
- **Research Focus:** Engineered an integrated computational platform combining Artificial Intelligence (AI) and Molecular Dynamics (MD) simulations to analyse and predict the structural and functional behaviours of 'click' hydrogels.

Junior Research Fellow | DST-Sponsored Project |Sept 2021 – Dec 2023

- **Project:** Sustainable, Biodegradable, and Affordable Substitutes for 'Single-Use Plastics'
- **Research Focus:** Researched and developed eco-friendly, biodegradable material alternatives to single-use plastics by formulating novel composites using castor oil and agricultural stubble aggregates.